

ZeroToken Router

Hybrid Token-Efficient Routing Agent — Track 1

AMD Developer Hackathon: ACT II

0 Fireworks tokens · every task
answered

factual

math

sentiment

summarize

NER

debug

logic

codegen

The leaderboard is one number: Fireworks tokens

Track 1 ranks submissions that pass the accuracy gate by total tokens, ascending. The rules say local inference inside the container counts fully toward accuracy — and zero toward tokens.

10 min

runtime cap on the grading VM

**4 GB / 2
vCPU**

grading environment

0

tokens is the optimal — and legal
— score

So the winning router is the one that (almost) never has to leave the box.

Architecture: local-first cascade

1

Regex classifier

8 categories, deterministic, microseconds, free

2

Qwen3.5-2B local (llama.cpp, CPU)

4-bit, 1.3 GB — answers everything · 0 tokens

3

Fireworks safety net

smallest ALLOWED_MODELS entry, only on failure or time pressure

Thinking mode ON only for math & logic · OFF elsewhere — controlled per request via the chat template

Engineered to never fail the harness

Adaptive budgets

Measures real tokens/sec after a warm-up request; sizes every generation to finish inside the 10-minute cap.

Easy-first ordering

No-think categories run first — if time gets tight, only reasoning-heavy tasks see smaller budgets.

Answer salvage

If a response is cut mid-reasoning, the agent retries without thinking, then falls back to the thought tail.

Code self-check

Generated Python is AST-parsed; syntax failures trigger one free local retry.

Always valid output

results.json is always written, always valid JSON, exit code always 0.

Env-driven config

FIREWORKS_API_KEY / BASE_URL / ALLOWED_MODELS read at runtime — nothing hardcoded.

What the judge sees

Factual

“The capital of Australia is Canberra, ... near Lake Burley Griffin.”

Sentiment

“Mixed; balances positive praise for battery life with negative criticism of the screen.”

Summarization

One-sentence constraint followed exactly.

NER

Maria Sanchez – Person · Fireworks AI – Organization · Berlin – Location · March – Date

Code debug

Bug identified in one sentence + corrected implementation.

Fireworks tokens spent: **0**

Built lean, ships anywhere

Model Qwen3.5-2B (Apache-2.0), Unsloth dynamic Q4_K_XL GGUF

Serving llama.cpp server — official image, OpenAI-compatible API

Agent Single-file Python 3, stdlib only, zero pip dependencies

Image linux/amd64 · ~2 GB compressed (10 GB limit)

ZeroToken Router — pass the gate, spend nothing, top the board.